

Muhammad Rafi Arsyia

Software Engineering Student

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Profile

Software Engineering student at Universiti Malaya building full-stack and AI applications with Next.js, Node.js and Python, self-hosted and containerised on a personal Linux server behind a Cloudflare Tunnel. Combines infrastructure work (Docker, NGINX, PM2) with applied AI/ML (RAG, CNNs written from scratch) and independent product building. Seeking a Data Engineering or Software Engineering internship.

Education

Universiti Malaya (UM), Malaysia

Since 2025

B.Sc. Computer Science (Software Engineering)

- Coursework: Data Structures & Algorithms, Machine Learning, Network Technology, Human Computer Interaction, Computer Systems Architecture.

International Program ICBS, Riau, Indonesia

2022 to 2025

Senior High School

- International curriculum with English-medium instruction.
- Active in student leadership and language programs.

Experience

PEKOM: Computer Science Society, Universiti Malaya

Since 2026

Technical Executive Committee

- Serving in the Website Division, building and maintaining the society's web platforms.
- Supporting faculty-wide events across the Faculty of Computer Science & IT.

PPI Malaysia (Indonesian Students Association)

Since 2025

Head of Website Division, Data & Information System Bureau

- Full ownership of the official PPI Malaysia website: development, maintenance, and continuous improvement.
- Serving Indonesian students across Malaysia.

PPI University of Malaya

Since 2026

Head of Department, Art Exhibition at IDFEST

- Directing the Art Exhibition division for IDFEST, an international event showcasing Indonesian heritage at Universiti Malaya, from concept and curation through to delivery.
- Leading the department end to end: theme and curation, artist and submission coordination, exhibition layout, and delivery on event day.

PPI University of Malaya

Since Jun 2026

Website Development Committee at IDFEST

- Building the official IDFEST website with Next.js 16, React 19 and Tailwind CSS v4, translating Figma designs into production-ready pages.
- Owning the homepage sections and animated timeline, hero responsiveness across breakpoints, and shared navigation and mascot components reused by the Art Exhibition and Musical Theater pages.

Volunteering

Faculty of Computer Science & Information Technology, Universiti Malaya

1 to 5 September 2026

Student Buddy, Inbound Mobility Programme 2026

- One of a four-person committee running a five-day FSKTM programme hosting students from Chulalongkorn University, Thailand's top-ranked university, in Kuala Lumpur.
- Served as a Student Buddy to the visiting delegation and was responsible for documentation across the whole programme, covering sessions and daily activities on all five days.

LARAS, PPI Malaysia

2026

Transportation Team & Field Committee

- Handled logistics, on-ground coordination and event-day operations for LARAS, PPI Malaysia's annual gathering.

Selected Projects

PaperMind: Local RAG Paper Analyzer

2026

Ask questions of any PDF and get grounded answers from a local LLM. FastAPI backend, ChromaDB vector store, Ollama (phi3:mini) inference. Fully offline with zero external API calls, containerised with Docker Compose and self-hosted on a Linux mini PC behind a Cloudflare Tunnel.

AI Digit Recognizer: CNN Built From Scratch

2026

Convolutional network written entirely in NumPy: im2col-vectorized convolutions, max-pooling, backpropagation and Adam, with no PyTorch and no TensorFlow. 99.1% test accuracy on MNIST, served by Flask with a visualizer rendering per-pixel saliency and live signal flow through every layer.

ThoughtLog: Full-Stack Blog with Custom CMS

2026

Node.js, Express and PostgreSQL with Markdown authoring and a three-state auto-publishing scheduler. Full-text search on tsvector/GIN with tag filtering, RSS and email subscriptions. Self-hosted behind a Cloudflare Tunnel, PM2-managed, deployed via GitHub Actions.

Steam Market Intelligence

2026

In-browser analytics over a 1,400-title dataset running real SQLite compiled to WebAssembly, with a live query console. Window functions and recursive CTEs execute client-side, so there is no backend and no precomputed export.

snip: Serverless URL Shortener

2026

API Gateway, Lambda and DynamoDB defined as infrastructure-as-code with AWS SAM for one-command provisioning. No managed server, least-privilege IAM per function, scales to zero when idle.

Crop Disease Detector

2026

Leaf-photo classifier across 38 disease categories trained on PlantVillage (54K+ images). MobileNetV2 transfer learning with data augmentation and two-phase fine-tuning in TensorFlow/Keras, deployed to Hugging Face Spaces with top-3 confidence scores.

IDFEST 2026 Website

2026

Official site for an international cultural event at Universiti Malaya, built with a committee. Next.js 16 static export, React 19 and Tailwind CSS v4. Owned the homepage timeline, hero responsiveness across breakpoints, and the shared navigation and mascot components reused across sections.

Arena Duel

2026

Server-authoritative real-time 1v1 over WebSockets. Simultaneous blind actions resolved deterministically server-side, with state machines, cooldowns and disconnect handling, so the client is never trusted with the outcome.

Skills

Languages: TypeScript, JavaScript (ES6+), Python, Java, PHP, SQL

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express, FastAPI, Flask, REST APIs, Socket.IO

Databases: PostgreSQL, MySQL, SQLite, Redis, Supabase

Infrastructure & Cloud: Docker & Compose, NGINX, Ubuntu self-hosting, Cloudflare Tunnel & Pages, Git, Bash, AWS (Lambda, DynamoDB, API Gateway, SAM)

AI / ML & Data: NumPy, TensorFlow, Keras, OpenCV, MediaPipe, pandas, scikit-learn, Retrieval-Augmented Generation (ChromaDB, Ollama)

Certifications & Courses

- Ultimate AWS Certified Developer Associate (DVA-C02), Udemy course completion, Stephane Maarek, June 2026
- Ultimate AWS Certified AI Practitioner (AIF-C01), Udemy course completion, Stephane Maarek, May 2026
- IELTS Academic, Band 5.5 (CEFR B2), British Council IDP, June 2025
- Additional coursework, around 570 hours across 2025 and 2026: freeCodeCamp JavaScript Algorithms & Data Structures (300 hours), The Complete JavaScript Course (Jonas Schmedtmann), full-stack bootcamps with Angela Yu and Cloudaffle, Advanced SQL (Maven Analytics), The Complete SQL Bootcamp (Jose Portilla), Spring AI Fast Track, and Cloud & Gen AI on AWS plus Spec-Driven Development with Kiro (Dicoding)

Languages

Bahasa Indonesia: Native | English: Professional working proficiency | Malay: Professional working proficiency